Ayush N P

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PROFESSIONAL SUMMARY

I am a prefinal-year Computer Science Engineering student with a strong passion for Data Science and Machine Learning, and a keen interest in understanding how systems and networks work beneath the surface. I enjoy hands-on learning through practical projects that connect theory with real-world applications. With a mindset focused on continuous improvement, I actively upskill myself through self-learning and experimentation. I am driven by curiosity and a desire to build reliable, data-driven solutions that create meaningful impact

EDUCATION

Vidyavardhaka College of Engineering, 2023 – 2027 | 9.29 CGPA
Bachelor of Computer Science Engineering
Relevant Course: Data Structures, Analysis and Design of Algorithms, Object oriented Programming, Database Management System, Computer Networks, Operating Systems, Basics of AI, DevOps, Machine Learning, Deep Learning

SKILLS

Programming Languages
Python, C, Java, SQL

Frameworks and Tools
Numpy, Pandas, MySQL, Git, GitHub, Django, FastAPI, Streamlit, Figma, Scikit-learn, Hugging Face, Jenkins , Docker , Kubernetes

AWARDS

2nd Place in Hackmitten 2.0 Hackathon, *MIT* 2025
Developed an AI-driven web application for pronunciation improvement using React, FastAPI, and the Wav2Vec2 model to analyze speech and provide real-time feedback.

2nd Place in Advanced Python and Machine Learning competition, *DEVCOM VVCE* 2025
Applied insights from *The Psychology of Money* to develop a machine learning–based model that provides personalized financial advice

2nd Place in a DevOps Hackathon, *VVCE* 2025
Built a secure and automated CI/CD pipeline using GitHub, Jenkins, Docker, and Slack, integrating security scans and auto-rollbacks to ensure reliable deployments.

EXPERIENCE

U&I, Mysore (NGO), *Teacher Volunteer* Nov 2025 – Present
Volunteering as a teacher with U&I, providing consistent educational support and mentorship to students through engaging lessons, while fostering a positive and encouraging learning environment.

PROJECTS AND CERTIFICATIONS

AI-Powered Pronunciation Coach, *(Project)*
Designed and developed an AI-driven web application to help users improve pronunciation by analyzing speech input and providing real-time feedback. Built an interactive React frontend with a FastAPI backend, leveraging the Wav2Vec2 pre-trained model to evaluate pronunciation accuracy and deliver meaningful performance insights

Multi-Threaded FTP Server with Client–Server Architecture, *(Project)*
Designed and implemented a custom FTP-like system using Python TCP sockets. Built separate control and data connections with dynamic port allocation, supported concurrent clients using a thread-per-client model, and implemented secure file upload/download with synchronization. Strengthened understanding of Computer Networks and Operating Systems concepts including TCP/IP, socket programming, concurrency, and file systems

Automated DevSecOps Pipeline with Self-Healing Rollbacks, *(Project)*
Designed and implemented a robust Jenkins and Docker-based CI/CD pipeline that automates the full software delivery lifecycle, from code checkout to production deployment. The system prioritizes security by integrating Gitleaks for automated secret detection and Trivy for multi-stage vulnerability scanning of both the filesystem and Docker images. To ensure high availability, I engineered a self-healing Auto-Rollback mechanism that monitors container health and automatically reverts to the last stable image upon detecting a deployment failure.

Python for Everybody — University of Michigan, *Coursera*
Developed a strong foundation in Python programming and problem-solving.

Operating Systems Fundamentals — Akamai Technologies, Inc. , *Coursera*
Gained a strong understanding of core operating system concepts, including processes, memory, and scheduling.

Introduction to DevOps — IBM, *Coursera*
Gained foundational knowledge of DevOps practices, including CI/CD, automation, and collaborative workflows.